

Introduction

In this project, we applied transfer learning by fine-tuning the pretrained GPT-2 model on a new corpus composed of movie dialogues, titles, release years, genres, and character names. The objective was to develop a chatbot with specialized knowledge of movies and their dialogues, capable of engaging in conversations about films. The original data, provided as text files, underwent standard preprocessing techniques to ensure it was properly cleaned, merged, and formatted for tokenization, making it ready for fine-tuning on the pretrained GPT-2 model.

Text Preprocessing

The data preprocessing involved several key steps. Special characters were removed using regular expressions to eliminate noise from the text. Specifically, the pattern `[^a-zA-Z0-9 ]+` was used to replace non-alphanumeric characters with empty spaces, ensuring the data was clean and standardized. The text was further normalized by converting all characters to lowercase to maintain consistency. Additionally, the original text files—`movie_lines.txt` and `movie_titles_metadata.txt`—were separated by a custom delimiter (`' +++$+++ '`), which required restructuring the files into a unified format suitable for training. The result of this process was a new text file containing clean, normalized, and well-structured data.

One of the challenges in natural language processing projects is the computational resources required for model training. Training large datasets can be resource-intensive, particularly when working with extensive corpora. Given the limited computational resources available for this project, we needed to reduce the dataset size to a more manageable level. To achieve this, we sampled 50% of the original dataset, balancing the need for representative data while ensuring the model could be trained efficiently. This downsized dataset still retained the essential structure and variety of the original corpus, enabling the model to learn effectively from the movie-related content while respecting the computational constraints.

Model Architecture: Transformer-Based Generative Models

The chatbot leverages the Transformer architecture, specifically GPT-2, which is well-suited for natural language generation tasks. The Transformer model, introduced by Vaswani et al. (2017), has become the standard for most natural language processing applications due to its ability to handle long-range dependencies and parallelize computations. Unlike traditional recurrent networks such as RNNs or LSTMs, the Transformer uses a self-attention mechanism that allows it to efficiently process entire sequences without relying on sequential order. GPT-2, as an

extension of the Transformer, stacks multiple layers of these self-attention mechanisms, enhancing its ability to generate context-aware responses (Radford et al., 2019).

A crucial component of this architecture is the input embedding layer, which transforms tokenized text into dense vector representations. This project uses the GPT-2 tokenizer, which applies Byte Pair Encoding (BPE) to break down input text into subword units, making it more efficient in handling rare or complex words (Sennrich et al., 2016). This process allows the model to capture semantic meaning at both word and subword levels, improving its understanding of language. When a user inputs the phrase “Hello, how are you?”, the tokenizer converts it into the following sequence of token IDs. These token IDs are passed to the model as input, ensuring that the raw text is efficiently represented in numerical form.

Since Transformers do not have a sense of word order, positional encodings are added to the input embeddings to provide the model with a sense of sequence (Vaswani et al., 2017). These encodings allow the model to understand the relative position of tokens within a sentence, which is essential for generating grammatically coherent text. In GPT-2, positional encodings are added directly to the word embeddings, enabling the model to treat sequential information explicitly during self-attention operations. For instance, the embedding of the first token in a sequence will be modified differently from the second token through position-specific vectors, helping the model distinguish their respective roles. This approach ensures that even though the model processes input in parallel, it retains the necessary information about the order of words, leading to coherent and contextually appropriate responses.

By combining input embeddings with positional encodings, GPT-2 effectively captures the semantic meaning of words while maintaining the correct word order, enabling it to generate natural, human-like dialogue. These foundational components are essential to the model’s ability to perform well in fine-tuned applications, such as a movie dialogue chatbot. In the following sections, we will go into the construction of these embeddings and discuss the challenges encountered during the development process. This includes the complexities of building the model architecture, training the model efficiently with limited resources, fine-tuning the model, exploring prompt-engineering strategies and overcoming difficulties in creating a coherent chatbot that can respond as an expert in movie dialogues.

Model Performance and Evaluation

The performance of the GPT-2 based movie chatbot was satisfactory, but with noticeable limitations. While it was able to engage in conversations about movies, characters, and genres due to the specialized fine-tuning on a movie-related dataset, its responses were not always consistent or deeply contextual. The chatbot often generated relevant, but sometimes repetitive, replies to movie-related prompts. This behavior can be attributed to the nature of GPT-2, which, despite fine-tuning, tends to rely on pattern recognition rather than a comprehensive understanding of context. As an autoregressive model, GPT-2 predicts each token based on

previous tokens, which can lead to a cascade of errors during extended conversations. This issue, coupled with its limited ability to retain long-term context, caused some of the responses to feel disjointed or irrelevant, particularly during longer exchanges.

Several factors contributed to the chatbot's performance. First, the dataset size, which was reduced for computational efficiency, likely limited the chatbot's ability to generalize across a broader spectrum of movie-related content. Furthermore, the model was trained for only one epoch, which, while sufficient for initial domain adaptation, may not have provided enough time for the model to fully capture the intricacies of the dialogue patterns. The GPT-2 architecture itself also plays a role, as it is designed to generate text based on probabilistic patterns, making it less adept at handling deep reasoning and maintaining context over long conversations.

To improve the chatbot's performance, several strategies could be considered. Expanding the dataset to include a wider variety of movie-related content, such as plot summaries or reviews, could help the chatbot develop a more comprehensive understanding of film discussions. Additionally, increasing the number of training epochs, coupled with early stopping to prevent overfitting, might enhance the model's ability to generalize and produce coherent responses. Incorporating a memory mechanism that allows the chatbot to retain context over longer exchanges would also significantly improve conversational continuity. Another potential enhancement is the integration of a hybrid model that combines GPT-2's generative capabilities with a retrieval-based system for accessing real-time information about movies, which could help generate more accurate and fact-based responses. Lastly, upgrading to a more advanced language model like GPT-3 could dramatically improve fluency and contextual understanding, albeit with increased computational demands.

Key Insights and Lessons Learned

Through this project, we gained valuable insights into the challenges and intricacies of developing a specialized chatbot using transfer learning with GPT-2. One of the key lessons learned was the importance of dataset quality and diversity. Although the model was fine-tuned on a rich corpus of movie-related data, the reduced dataset size highlighted the trade-offs between computational efficiency and the chatbot's ability to generalize across a broad spectrum of topics. We also observed the limitations of GPT-2 in maintaining long-term context during conversations, particularly due to its autoregressive nature, which led to some disjointed responses in extended dialogues. Moreover, we learned that careful tuning of hyperparameters, such as temperature and top-p sampling, is essential for balancing response creativity and coherence. The project also emphasized the significance of training duration, shorter training times limit the model's potential, while more epochs and early stopping could greatly enhance performance. Finally, this project reinforced the idea that incorporating external mechanisms, such as dialogue memory or hybrid models, can significantly improve the chatbot's

conversational quality and factual accuracy. These insights will inform future work, especially in designing more robust and context-aware dialogue systems.

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